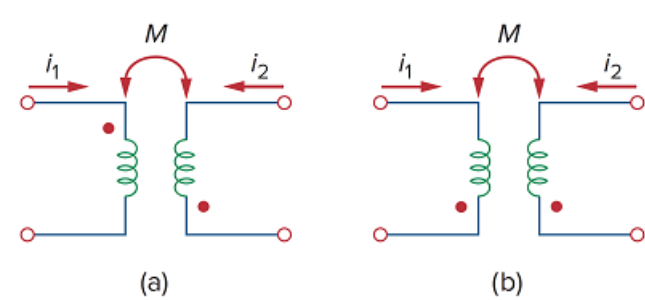


Problem

For the two magnetically coupled coils of Fig. 13.69(b), the polarity of the mutual voltage is:

- (a) Positive
- (b) Negative

Figure 13.69



Step-by-step solution

Step 1 of 2

Refer to Figure 13.69 (b) in the textbook.

The reference polarity of the mutual voltage is depends on the current through the dotted terminal of the coil.

Step 2 of 2

If a current enters through the dotted terminal, the polarity of the secondary voltage is positive otherwise negative.

From the Figure, the current enters through the dotted terminal, the polarity of the secondary voltage is positive.

Thus, the correct answer is **a**.